

AMT Horizon

Functional Specification Document

DOCUMENT VERSION 2.0

16/01/2026

Author

Name	Role
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Revision History

Date	Version	Description of Changes
04/01/2026	1.0	- Document creation and drafting of sections Overview, A to E. - Added Functional Requirements.
08/01/2026	1.1	- Translated code and technical sections to English. - Adapted for NextJS, Vitest, Neon, Drizzle, better-auth.
14/01/2026	1.2	- Separated Functional and Technical Specifications into distinct documents. - Added Page Functionality section. - Updated Functional Requirements with detailed descriptions.
16/01/2026	2.0	Restructured Functional Requirements into clear, non-technical lists grouped by functional areas (User Management, Ride Management, etc.). Added Two-Factor Authentication to the login flow. Improved Mermaid diagrams for clarity and consistency, including error handling and validation steps. Enhanced Personas section with textual summaries alongside mindmaps. Removed redundant content between Product Functional Capabilities and Functional Requirements. Added GDPR compliance notes to relevant flows (e.g., account deactivation, data export). Standardized terminology across the document. Added notes for edge cases (e.g., OCR failure, validation errors) directly in diagrams where applicable.

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I. General Overview

Client Request: A **web and mobile application** to manage taxi rides, driver assignments, and billing. The goal is to provide an **intuitive, all-in-one solution** for admin, drivers, and customers, featuring **automatic ride data import** and **smart ride suggestions**.

AMT Horizon is designed to **simplify daily operations** for taxi companies by replacing manual processes with a streamlined digital system. The application helps:

- **Reduce errors** through automation of repetitive tasks.
- **Save time** with real-time scheduling and tracking tools.
- **Improve satisfaction** for both drivers and customers with a smooth, transparent experience.

A. Product Description

AMT Horizon is a **complete taxi management platform** that enables:

- **Centralized recording and organization** of taxi rides.
- **Automatic ride distribution** to drivers based on their **availability and location**.
- **Invoice generation** (individual or grouped) with **automatic payment reminders** for overdue bills.
- **Smart ride suggestions** for drivers, allowing them to accept or request assignments based on their preferences.
- **Easy data import** from forms, text, or even **photos of tickets or invoices** (no manual entry required).

This solution meets the needs of:

- **Fleet managers**, who need a **clear dashboard** to oversee operations and generate reports.
- **Drivers**, who want to **maximize earnings** and manage their trips flexibly.
- **Customers**, who expect **easy booking, real-time tracking**, and **secure payment options**.

B. Product Functional Capabilities

- **User Management:** [See II.A.1](#)
- **Ride Management:** [See II.A.2](#)
- **Admin Dashboard:** [See II.A.3](#)
- **Billing and Payments:** [See II.A.4](#)
- **Notifications:** [See II.A.5](#)
- **Additional Features:** [See II.A.6](#)

C. Project Organization

Project Representatives

Project Owner	Represented by...
Client	Alexis SANTOS (PM)

Stakeholders

Stakeholder	Key Interests
Admins	Full ride and invoice management, reporting.
Drivers	Access to suggested rides management.
Customers	Easy booking, real-time tracking, secure payments.

Voici une version **réorganisée et simplifiée** de la section **II. Requirements**, adaptée pour un public non technique tout en conservant la clarté et la structure. J'ai transformé les tableaux en **listes structurées** et regroupé les fonctionnalités par thème logique, avec des descriptions en langage naturel.

II. Requirements

A. Functional Requirements

1. User Management

Objective: Ensure secure and efficient management of user accounts for all roles (Admins, Drivers, Customers).

- Account Creation and Login**
Users can create accounts and log in securely using their email and password. The system supports three roles: Admin, Driver, and Customer.
(Detailed flow: [IV.A. Login Page](#))
- Password Recovery**
Users can reset their password via a secure link sent to their email, valid for 24 hours.
- Driver Account Validation**
Admins manually verify driver accounts using submitted documents (ID card, driver’s license) within 24 hours.
- Profile Updates**
Users can update their personal information (email, phone number) with validation to ensure data accuracy.
- Account Deactivation**
Admins can deactivate user accounts, with optional reasons provided. Deactivated accounts are archived in compliance with data protection regulations.
(Detailed flow: [IV.G. Account Deactivation](#))

2. Ride Management

Objective: Streamline the process of creating, assigning, and tracking rides.

- **Manual Ride Entry**

Users add rides via a form (date, time, location, price).

(Detailed flow: [IV.B.1 Ride Management Flow](#))

- **Automatic Ride Import**

Admins/drivers upload ticket/invoice photos. System extracts data automatically with manual validation.

(Detailed flow: [IV.H. OCR Import Page](#))

[!Note] If OCR fails, users manually correct data before saving.

- **Ride Suggestions**

System suggests rides based on driver availability/location.

(Detailed flow: [IV.C.1 View Suggested Rides Flow](#))

- **Ride Assignment Requests**

Drivers request assignments (max 5 simultaneous requests).

- **Ride History**

Drivers and customers view past rides (each user sees only their own rides).

(Detailed flow: [IV.D.2 Track Rides Flow](#))

- **Ride Cancellation**

Customers and admins can cancel rides up to 2 hours before departure with a mandatory reason.

- **Driver Ratings**

Customers rate drivers (1-5 stars) post-ride.

(Detailed flow: [IV.D.5 Rate Drivers Flow](#))

3. Admin Tools

Objective: Provide admins with powerful tools to manage rides, drivers, and data.

- **Dynamic Ride Table** Admins view, sort, filter, and export ride data (max 10,000 rows).

(Detailed flow: [IV.I. Admin Excel-like View](#))

- **Statistics and Reports** System generates daily charts/reports on rides, revenue, and performance.

4. Billing and Payments

Objective: Automate and simplify the billing process for admins and customers.

- **Invoice Creation** Admins can generate individual or grouped invoices automatically, with unique numbering.

(Detailed flow: [IV.E.1 Invoice Generation Flow](#))

- **Payment Reminders** The system sends automatic email reminders for overdue payments at set intervals (D+14, D+21, and D+30 after the due date).
(Detailed flow: [IV.E.2 Payment Tracking Flow](#))
 - **Discounts** Admins can apply exceptional discounts to invoices, with mandatory justification.
(Detailed flow: [IV.E.3 Discounts Flow](#))
 - **Invoice Delivery** Invoices can be sent directly through the app, with PDF attachments and status updates.
(Detailed flow: [IV.E.1 Invoice Generation Flow](#))
 - **Partial Payments** Customers can make partial payments (minimum 30% of the total amount).
-

5. Notifications

Objective: Keep all users informed with real-time alerts and customizable notifications.

- **Real-Time Alerts** Users receive notifications for new rides, assignments, and invoices via email or in-app alerts.
 - **Customizable Channels** Users can choose their preferred notification method (email, in-app, or push notifications).
 - **Urgent Notifications** The system sends immediate alerts for critical events, such as ride cancellations by customers.
-

6. Additional Features

Objective: Enhance the user experience with advanced tools and integrations.

- **Geolocation and Maps** Drivers and customers can view ride locations and track routes in real-time using integrated maps.
(Detailed flow: [IV.K. Geolocation & Maps](#))
 - **In-App Chat** Drivers and customers can communicate directly through a built-in chat, with a 30-day message history.
(Detailed flow: [IV.J. Real-Time Chat](#))
 - **Calendar Integration** Drivers can sync their ride schedules with personal calendars (Google Calendar, Outlook).
(Detailed flow: [IV.L. Calendar Integration](#))
 - **Activity Logs** Admins can view logs of user actions, retained for 6 months in compliance with GDPR.
 - **Two-Factor Authentication** Two-factor authentication is mandatory for admins and optional for other users for enhanced security.
(Detailed flow: [IV.A. Login Page](#))
-

B. Non-Functional Requirements

- **OCR Accuracy:** The system must accurately extract data from photos or documents with at least **90% precision**.
- **Performance:** Critical actions (e.g., ride assignment, payment processing) must complete in **under 2 seconds** to ensure a smooth user experience.
- **Security:** All user data must be encrypted and stored in compliance with **GDPR and other data protection regulations**.
- **Support Response Time:** Customer support must respond to inquiries within **24 hours** during business days.

III. Personas

A. Business Manager (Marc, 55, Schedule Management)

Manager of a small taxi fleet in Île-de-France

```
mindmap
  root("**Business Manager (Marc, 55)**")
    Role
      Manager of a small taxi fleet in Île-de-France
    Main Goals
      Zero scheduling errors
      Cover peak demand
      Reduce manual work
    Key Needs
      Simple & visual tool
      Real-time availability overview
      Quick ride assignment
      Alerts for cancellations
      Driver performance tracking
    Frustrations
      3-4h/day manual scheduling
      Unresponsive drivers
      High turnover
    Feature Impact
      Excel-like admin view
      Push notifications + SMS
```

B. Administrative Secretary (Nathalie, 47, Billing Efficiency)

```
mindmap
  root("**Administrative Secretary (Nathalie, 47)**")
```

Role

Administrative secretary in a transport SME

Main Goals

Send invoices on time without errors

Track overdue payments effortlessly

Centralize documents to avoid misplacement

Key Needs

Automated system

Generate & send invoices in one click

Track payments & send automatic reminders

Store contracts, invoices, purchase orders in one place

Clear & intuitive interface

Frustrations

Lost tickets = incomplete invoices

20% late payments affect cash flow

Time wasted searching for documents

Feature Impact

OCR integration for invoice generation & cloud archiving

Automated payment reminders (D+14, D+21, D+30)

Unified document management with search functionality

C. Driver (Djamel, 39, Income Maximizer)

```
mindmap
```

```
  root("**Driver  
(Djamel, 39)**")
```

Role

Independent taxi driver

Main Goals

Maximize earnings with minimal effort

Choose the most profitable rides

Avoid problematic customers

Key Needs

App that allows him to

See available rides with price and distance before accepting

Request assignment for preferred rides

Receive notifications for high-paying rides

View ride history and earnings

Frustrations

Low-paying rides or last-minute route changes

Lack of transparency on fares

Having to call dispatch for ride details

Feature Impact

Algorithm prioritizing high-revenue and long-distance rides

Real-time ride details and fare transparency

D. Classic Customer (Jean-Michel, 62, Reliability Focus)


```

mindmap
  root("**Classic Customer  
(Jean-Michel, 62)**")
    Role
      Retired, regular taxi user.
    Main Goals
      Find a taxi **quickly and without complications**.
      Have a **punctual and professional** driver.
      Know the exact fare upfront.
      Pay by cash or card without issues.
    Key Needs
      A **simple and reliable** solution to:
        Book a taxi by phone or via a basic app.
        Get a clear fare estimate before the ride.
        See driver photo and rating before acceptance.
      A **consistent and polite** service.
    Frustrations
      Long wait times during peak hours.
      Surprise fees or unclear pricing.
      Drivers unfamiliar with optimal routes.
    Feature Impact
      Locked-in fare at booking.
      Driver rating system.
      Real-time driver tracking.

```

E. Corporate Customer (Claire, 41, Team Ride Planning)

```

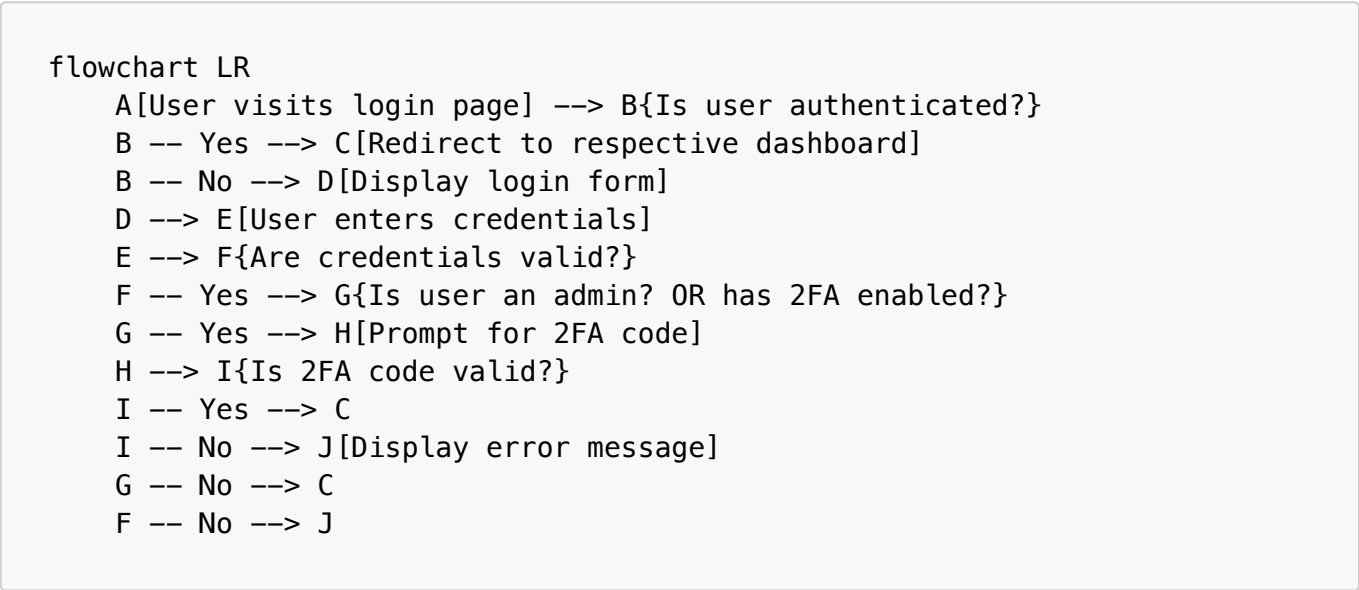
mindmap
  root("**Corporate Customer  
(Claire, 41)**")
    Role
      Executive assistant managing travel for XX employees.
    Main Goals
      Plan rides for teams without fail.
      Track ride progress in real time.
      Have **direct contact with drivers** in case of issues.
    Key Needs
      A platform to:
        Book multiple taxis at once for group travel.
        Receive **real-time notifications** for delays or cancellations.
        Contact drivers directly for last-minute changes.
      **Clear reporting** to justify expenses.
    Frustrations
      Drivers arriving late, causing employees to wait.
      Lack of visibility on ride status.
      Invoices arriving late or with errors.
    Feature Impact
      Real-time GPS tracking with delay alerts.
      In-app chat for direct driver contact.
      Grouped invoice generation for multiple rides.

```

IV. Page Functionality

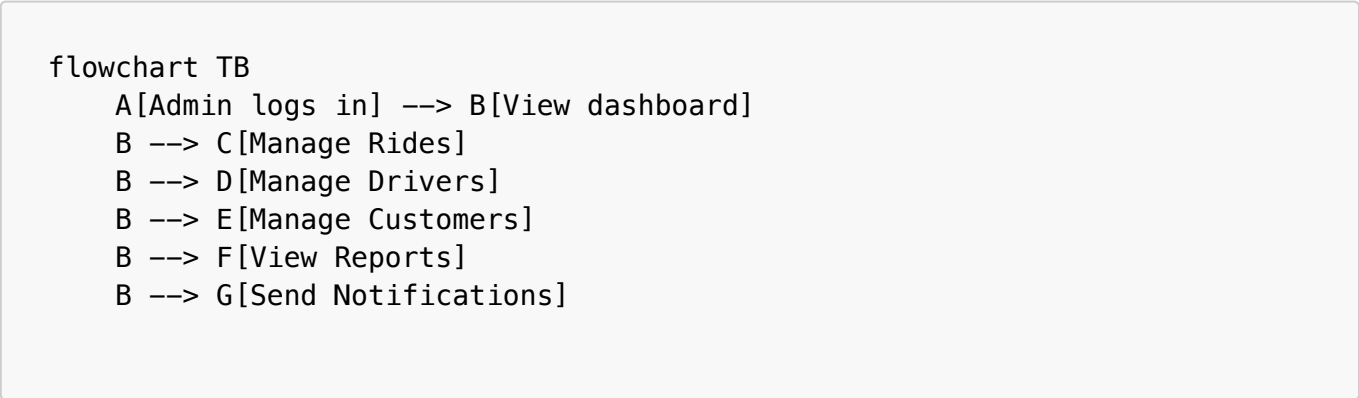
A. Login Page

- **Purpose:** Authenticate users and redirect them to their respective dashboards.
- **Actors:** All users.
- **Features:**



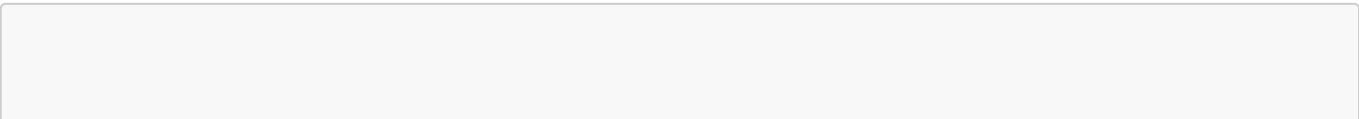
B. Admin Dashboard

- **Purpose:** Central hub for managing rides, drivers, customers, and invoices.
- **Actors:** Admins.
- **Features:**



1. Ride Management Flow

- **Purpose:** Add, edit, delete, assign rides, and import via OCR.



flowchart LR

```

    A["Admin in 'Manage Rides'"] --> B[View ride list]
    B --> C{Action?}
    C -- Add --> D[Open ride form] --> E[Save ride]
    C -- Edit --> F[Select ride] --> G[Modify details] --> E
    C -- Delete --> H[Select ride] --> I[Confirm deletion] --> J["Archive ride (GDPR)"]
    C -- Assign --> K[Select ride] --> L[Choose driver] --> M[Confirm] --> E
    C -- Import --> N[Upload photo/text] --> O[OCR extract]
    O --> P{Data valid?}
    P -- Yes --> E
    P -- No --> Q[Manually correct] --> O
  
```

[!Note]

- Deleted rides are archived for 6 months (GDPR compliance).
- OCR errors route to manual correction.

2. Manage Drivers Flow

- **Purpose:** Add, edit, deactivate drivers, and validate documents.

flowchart LR

```

    A["Admin in 'Manage Drivers'"] --> B[View driver list]
    B --> C{Action?}
    C -- Add --> D[Open form] --> E[Save driver] --> P
    C -- Edit --> F[Select driver] --> G[Modify details] --> E
    C -- Deactivate --> H[Select driver] --> I[Confirm] --> J[Deactivate account] --> P
    C -- Validate Docs --> K[Review documents] --> L{Approved?}
    L -- Yes --> M[Activate account] --> N[Notify driver]
    L -- No --> O[Reject with reason] --> N
    N --> P[Log action in history]
  
```

3. Manage Customers Flow

- **Purpose:** Add, edit, and deactivate customers.

flowchart LR

```

    A["Admin selects 'Manage Customers'"] --> B[View list of customers]
    B --> C{Select customer action}
    C -- Add Customer --> D[Open customer addition form]
    D --> E[Enter customer details]
    E --> F[Save customer]
    C -- Edit Customer --> G[Select customer to edit]
    G --> H[Modify customer details]
  
```

```
H --> F
F --> I[Log action in activity history]
```

4. View Reports & Statistics Flow

- **Purpose:** View ride statistics, financial reports, and export data.

```
flowchart LR
    A["Admin in 'View Reports & Statistics'"] -- Want to view Ride Statistics --> B["View charts (number of rides, revenue, driver performance)"]
    A -- Want to view Financial Reports --> C["View revenue trends, outstanding payments"]
    A -- Want to Export Data --> D["Select data to export"]
    D --> E["Click on 'Export Data' on Dashboard Data Card"]
    E --> F["Select file format (CSV/PDF)"]
    F --> G["Generate file and download"]
    G --> H["Log action in activity history"]
```

C. Driver Dashboard

- **Purpose:** View/manage rides, availability, earnings, and notifications.
- **Actors:** Drivers.
- **Features:**

```
flowchart TB
    A[Driver logs in] --> B[Dashboard Overview]
    B --> C[View Suggested Rides]
    B --> D[Manage Availability]
    B --> E[View Earnings & Ratings]
    B --> F[Access Notifications]
    B --> G[Open Chat]
```

1. View Suggested Rides Flow

- **Purpose:** View/request assignment for suggested rides.

```
flowchart LR
    A["Driver in 'Suggested Rides'"] --> B[View ride list]
    B --> C{Action?}
    C -- Request --> D[Click 'Request'] --> E[Send to admin]
    C -- View Details --> F[Select ride] --> G[Show details]
```

[!Note]

- Rides suggested based on location/availability.
- Max 5 simultaneous requests per driver.

2. Manage Availability Flow

- **Purpose:** Toggle availability status for ride suggestions.

```

flowchart LR
    A["Driver in Dashboard"] --> B[Locate availability toggle]
    B --> C{Toggle status}
    C -- Available --> D[Status: Available]
    C -- Unavailable --> E[Status: Unavailable]
    D --> F[Receive ride suggestions]
    E --> G[No new suggestions]

```

[!Note]

- Drivers only receive suggestions when marked as Available.
- Status updates are instantaneous.

3. Notifications Flow

- **Purpose:** View and manage notifications for rides, assignments, and messages.

```

flowchart LR
    A["Driver in Dashboard"] --> B[Click Notifications icon]
    B --> C[View notification list]
    C --> D{Action?}
    D -- Mark Read --> E[Update status]
    D -- Delete --> F[Confirm deletion]
    D -- Open --> G[View full details]

```

[!Note]

- Notifications include ride assignments, messages, and payment confirmations.
- Unread notifications are highlighted.

4. Earnings & Ratings Flow

- **Purpose:** View their earnings, completed rides, and customer ratings.

```

flowchart LR
    A["Driver in Dashboard"] --> B[Click Earnings & Ratings]
    B --> C[View earnings summary]
    B --> D[View completed rides]
    B --> E[View customer ratings]
    E --> F{Filter?}

```

```

F -- Yes --> G[Select time period] --> E
F -- No --> H[End]

```

[!Note]

- Earnings updated in real-time.
- Ratings include comments and 1-5 star scores.

D. Customer Dashboard

- **Purpose:** Book rides, track trips, manage payments, and rate drivers.
- **Actors:** Customers.
- **Features:**

flowchart TB

```

A[Customer logs in] --> B[View dashboard overview]
B --> C[Book a Ride]
B --> D[Track Rides]
B --> E[Manage Payments]
B --> F[Access Notifications]
B --> G[Rate Drivers]
B --> H[Contact Support]

```

1. Book a Ride Flow

- **Purpose:** Book rides, view estimated fares, and manage ride history.

flowchart LR

```

A["Customer in 'Book a Ride'"] --> B[Open form]
B --> C[Enter details] --> D[View fare]
D --> E{Confirm?}
E -- Yes --> F[Submit] --> G[Confirm booking] --> H[Notify driver]
E -- No --> B

```

[!Note]

- Fare locked at booking (no surprises).
- Driver receives real-time notification.

2. Track Rides Flow

- **Purpose:** Track assigned rides in real-time and view ride history.

flowchart LR

```

A["Customer in Track Rides"] --> B[View current ride status]

```

```

B --> C[Track driver on map]
B --> D[View ride history]
C --> E{Driver delayed?}
E -- Yes --> F[Show updated ETA]
E -- No --> G[Proceed as normal]

```

[!Note]

- Real-time GPS tracking with estimated arrival time.
- History includes receipts and driver ratings.

3. Manage Payments Flow

- **Purpose:** Manage ride payments, view invoices, and receive notifications.

flowchart LR

```

A["Customer in Manage Payments"] --> B[View rides with payment status]
B --> C{Action?}
C -- Pay --> D[Choose method: Card/Cash]
D -- Card --> E[Complete payment] --> F[Receive confirmation]
D -- Cash --> G[Driver marks as paid] --> F
C -- View Invoice --> H[Download PDF]

```

[!Note]

- Partial payments allowed (minimum 30%).
- Cash payments require driver confirmation in-app.

4. Notifications Flow

- **Purpose:** View and manage ride/status notifications.

flowchart LR

```

A["Customer in Dashboard"] --> B[Click Notifications]
B --> C[View list]
C --> D{Action?}
D -- Mark Read --> E[Update status]
D -- Open --> F[View details]
D -- Delete --> G[Confirm deletion]

```

[!Note]

- Notifications include **ride status**, **payment confirmations**, and **support messages**.
- Urgent alerts (e.g., cancellations) trigger **push notifications**.

5. Rate Drivers Flow

- **Purpose:** Rate drivers post-ride.

flowchart LR

```
A["Customer post-ride"] --> B[Notification to rate]
B --> C[Open rating form]
C --> D["Submit rating (1-5 stars)"]
D --> E[Add optional comment]
E --> F[Confirm submission]
F --> G[Driver receives feedback]
```

[!Note]

- Ratings are anonymous but linked to the ride
- Drivers can respond to comments via in-app chat
 - Warning for inappropriate comments.

6. Contact Support Flow

- **Purpose:** Contact admins via in-app chat for support.

flowchart LR

```
A["Customer in Dashboard"] --> B[Click Contact Support]
B --> C[Open chat interface]
C --> D[Send message]
D --> E[Admin responds]
E --> F[Conversation saved for 30 days]
```

[!Note]

- Support responses within 24 hours.
- Chat history archived for reference.

E. Billing Page (Admin)

- **Purpose:** Manage invoicing, payments, discounts, exports, and financial reporting.
- **Actors:** Admins.
- **Features:**

flowchart TB

```
A[Admin opens Billing Page] --> B[View invoicing options]
B --> C[Generate Invoices]
B --> D[Track Payments]
B --> E[Apply Discounts]
B --> F[Export Financial Data]
```


1. Invoice Generation Flow

- **Purpose:** Create individual or grouped invoices.

```
flowchart LR
    A["Admin in 'Generate Invoices'"] --> B[Select rides]
    B --> C{Type?}
    C -- Individual --> D[Generate PDF] --> E[Send invoice]
    C -- Grouped --> F[Select multiple rides] --> D
```

[!Note]

- Invoices auto-numbered.
- Sent via email/in-app.

2. Payment Tracking Flow

- **Purpose:** Track payment status and send automatic reminders for overdue invoices.

```
flowchart LR
    A["Admin in Track Payments"] --> B[View payment status list]
    B --> C{Action?}
    C -- Send Reminder --> D[Select overdue invoices]
    D --> E["Send reminder (D+14, D+21, D+30)"]
    C -- Update Status --> F[Select invoice] --> G[Mark as paid/cancelled]
    G --> H[Log action]
```

[!Note]

- Max 3 reminders per invoice.
- Status updates trigger customer notifications.

3. Discounts Flow

- **Purpose:** Apply exceptional discounts to invoices with justification.

```
flowchart LR
    A["Admin in Apply Discounts"] --> B[Select invoice]
    B --> C[Enter discount %]
    C --> D[Provide justification]
    D --> E[Apply discount]
    E --> F[Update invoice]
    F --> G[Notify customer]
```

[!Note]

- Justification archived for auditing.

4. Export Flow

- **Purpose:** Export invoices/financial reports to CSV/PDF.

flowchart LR

```
A["Admin in Export"] --> B[Select data range]
B --> G[Click on 'Export' button] --> C[Choose format: CSV/PDF]
C --> D[Generate file]
D --> E[Download]
E --> F[Log export in history]
```

[!Note]

- Max 10,000 rows per export.
- Exports include timestamp and admin ID.

F. Notifications Page (All Users)

- **Purpose:** Centralize notifications for rides, invoices, and alerts.
- **Actors:** All users.
- **Features:**

flowchart LR

```
A[User opens Notifications] --> B[View list]
B --> C{Action?}
C -- Mark Read --> D[Update status]
C -- Filter --> E[Select type] --> F[View filtered]
C -- Customize --> G[Open settings] --> H[Save preferences]
C -- View Details --> I[Open notification]
I -- Mark Read --> D
```

[!Note]

- Notifications retained for 30 days.
- Urgent alerts (e.g., cancellations) trigger push + email.

G. Profile Page (All Users)

- **Purpose:** View and edit personal information.
- **Actors:** All users.
- **Features:**

flowchart LR

```
A[User opens Profile] --> B[View personal info]
B --> C{Action?}
```

```

C -- Edit Info --> D[Update email/phone] --> E[Save changes] --> I
C -- Account Settings --> F[Toggle notifications]
C -- Deactivate --> G[Enter reason] --> H[Confirm] --> I["Archive data (GDPR)"]

```

[!Note]

- Email/phone updates require **verification**.
- Deactivated accounts are **archived for 6 months**.

H. OCR Import Page (Admin/Drivers)

- **Purpose:** Import ride data from photos or text using OCR.
- **Actors:** Admins, Drivers.
- **Features:**

flowchart LR

```

A[User opens OCR Import] --> B[Upload photo/text]
B --> C[OCR extract data]
C --> D{Data valid?}
D -- Yes --> E[Save ride]
D -- No --> F[Manually correct] --> D
E --> G[Notify user]

```

[!Note]

- Supports **handwritten tickets**.
- Manual correction required for **OCR errors**.

I. Admin Excel-like View

- **Purpose:** Dynamic table for bulk ride/driver management.
- **Actors:** Admins.
- **Features:**

flowchart LR

```

A[Admin opens Excel View] --> B[View dynamic table]
B --> C{Action?}
C -- Sort/Filter --> D[Apply filters]
C -- Bulk Actions --> E[Select rides/drivers] --> F{Action?}
F -- Assign --> G[Choose driver] --> H[Confirm]
F -- Delete --> I[Confirm] --> J[Archive data]
F -- Export --> K[Select format] --> L[Generate file]

```

[!Note]

- Bulk actions limited to 10,000 rows.
- Exports include metadata (timestamp, admin ID).

J. Real-Time Chat (Drivers/Customers)

- **Purpose:** Facilitate communication between drivers and customers.
- **Actors:** Drivers, Customers.
- **Features:**

flowchart LR

```
A[User opens Chat] --> B[View conversation list]
B --> C{Action?}
C -- Send Message --> D[Type message] --> E[Deliver]
C -- View History --> F[Show 30-day history]
```

[!Note]

- Messages **encrypted** for security.
- History retained for **30 days**.

K. Geolocation & Maps (Drivers/Customers)

For Drivers :

- **Purpose:** View ride location and share GPS in real-time.

flowchart LR

```
A["Driver opens Maps"] --> B[View ride locations]
B --> C[Enable GPS sharing]
C --> D[Customer sees location]
```

For Customers:

- **Purpose:** Track driver's location and estimated arrival time.

flowchart LR

```
A["Customer opens Maps"] --> B[Track driver]
B --> C[View ETA]
C --> D{Driver delayed?}
D -- Yes --> E[Update ETA]
D -- No --> F[Proceed]
```

[!Note]

- ETA updates **every 30 seconds**.
- Driver's photo/vehicle details **displayed**.

L. Calendar Integration

- **Purpose:** Sync ride schedules with personal calendars.
- **Actors:** Drivers.
- **Features:**

flowchart LR

A["Driver opens Calendar"] --> B[Link Google/Outlook]

B --> C[Sync rides as events]

C --> D[Receive alerts]